

# Java Programming Manual

---

Concepts, Algorithms, and Implementations

*A comprehensive guide covering Conditionals, Loops, Methods, and Object-Oriented Programming.*

# 1. Conditional Statements

## Question

Write a Java program to check whether a given integer is even or odd.

## Step-by-Step Process

1. Import the Scanner class to read input from the user.
2. Define the main class ConditionalState and the main method.
3. Prompt the user to enter an integer.
4. Read the integer using the `nextInt()` method.
5. Use an `if-else` statement to check the condition `n % 2 == 0`.
6. If true, print "the number is even", else print "the number is odd".

## Algorithm

1. **START**
2. Initialize a Scanner object to take user input.
3. Print "enter the number".
4. Read integer  $n$ .
5. If  $n$  modulo 2 is equal to 0, go to step 6, else go to step 7.
6. Print "the number is even". Go to step 8.
7. Print "the number is odd".
8. **STOP**

## Code Implementation

```
import java.util.Scanner;

public class ConditionalState {
    public static void main(String[] args) {
        Scanner eo = new Scanner(System.in);
        int n;
        System.out.println("enter the number");
        n = eo.nextInt();

        if(n % 2 == 0) {
            System.out.println("the number is even");
        } else {
            System.out.println("the numder is odd");
        }
    }
}
```

## Sample Output

```
enter the number
4
the number is even

enter the number
7
the numder is odd
```

## 2. Looping Statements

### Question

Write a Java program to generate and print the multiplication table of a given number.

### Step-by-Step Process

1. Import the Scanner utility.
2. Declare a class named LoopState.
3. Take an integer input  $n$  from the user.
4. Use a for loop starting from  $i = 1$  to  $i = 10$ .
5. In each iteration, calculate  $n * i$  and print the formatted string.

### Algorithm

1. **START**
2. Read integer  $n$ .
3. Print "the multiplication table of given number".
4. Initialize loop counter  $i = 1$ .
5. If  $i \leq 10$ , proceed to step 6. Else, go to step 9.
6. Calculate product =  $n * i$ .
7. Print  $n + "x" + i + " = " + product$ .
8. Increment  $i$  by 1 and go to step 5.
9. **STOP**

## Code Implementation

```
import java.util.Scanner;

public class LoopState {
    public static void main(String[] args) {
        Scanner m = new Scanner(System.in);
        int n;
        System.out.println("enter the number");
        n = m.nextInt();
        System.out.println("the multipilcation table of given number");

        for(int i = 1; i <= 10; i++) {
            System.out.println(n + "x" + i + " = " + (n*i));
        }
    }
}
```

## Sample Output

```
enter the number
5
the multipilcation table of given number
5x1 = 5
5x2 = 10
5x3 = 15
5x4 = 20
5x5 = 25
...
5x10 = 50
```

## 3. Methods

### Question

Write a Java program to calculate the factorial of a number using a user-defined static method.

### Step-by-Step Process

1. Create a static method `factorial(int n)` that returns an integer.
2. Inside the method, initialize a variable `fact = 1`.
3. Use a loop to multiply `fact` by loop variable `i` from 1 to `n`.
4. Return the calculated `fact` value.
5. In the main method, read input from the user and call `factorial(n)`.
6. Print the returned result.

### Algorithm

1. **START**
2. Read an integer `n`.
3. Call function `factorial(n)`.
4. Inside function, set `fact = 1`.
5. For `i = 1` to `n`, do `fact = fact * i`.
6. Return `fact`.
7. Print "factorial = " + result.
8. **STOP**

## Code Implementation

```
import java.util.Scanner;

public class MethodTask {
    static int factorial(int n) {
        int fact = 1;
        for(int i = 1; i <= n; i++) {
            fact = fact * i;
        }
        return fact;
    }

    public static void main(String[] args) {
        Scanner fa = new Scanner(System.in);
        int n, result;
        System.out.println("enter the number");
        n = fa.nextInt();

        result = factorial(n);
        System.out.println("factorial = " + result);
    }
}
```

## Sample Output

```
enter the number
5
factorial = 120
```

## 4. Object-Oriented Programming

### Question

Create a menu-driven Student Utility Program. Implement a `Student` class to manage student details, calculate total marks and percentage, and display the results.

### Step-by-Step Process

1. Create a `Student` class with attributes for name, roll number, subject marks, total, and percentage.
2. Implement `studentdetails()` to accept input from the console.
3. Implement `calc()` to process the total and percentage.
4. Implement `show()` to print the final data format.
5. In the `StudentProgram` main class, create a do-while loop to display a menu interface.
6. Use a switch case to route user choices to the corresponding `Student` methods.

### Algorithm

1. **START**
2. Create object `st` of class `Student`.
3. Display menu options (1. Enter Details, 2. Calculate, 3. Show, 4. Exit).
4. Read `choice`.
5. If choice is 1, call `st.studentdetails()`.
6. If choice is 2, call `st.calc()` (Total =  $s1+s2+s3$ , Percentage =  $\text{Total}/3.0$ ).
7. If choice is 3, call `st.show()`.
8. If choice is 4, print "exited" and break loop.
9. Repeat steps 3-8 while `choice != 4`.
10. **STOP**

### UML Class Diagram

The following structural representation highlights the attributes and behaviors of the `Student` class:

## Student

- studentname : String
- rollno : int
- s1, s2, s3 : int
- total : int
- percentage : double
- sp : Scanner

  

- + studentdetails() : void
- + calc() : void
- + show() : void

## Code Implementation

```

import java.util.Scanner;

class Student {
    String studentname;
    int rollno;
    int s1, s2, s3;
    int total;
    double percentage;
    Scanner sp = new Scanner(System.in);

    void studentdetails() {
        System.out.println("enter the name of student");
        studentname = sp.nextLine();
        System.out.println("enter th roll number");
        rollno = sp.nextInt();
        System.out.println("enter the firstsubject marks");
        s1 = sp.nextInt();
        System.out.println("enter the secondsubject marks");
        s2 = sp.nextInt();
        System.out.println("enter the thirdsubject marks");
        s3 = sp.nextInt();
        sp.nextLine(); // consume newline
    }

    void calc() {
        total = s1 + s2 + s3;
        System.out.println("total is " + total);
        percentage = total / 3.00;
        System.out.println("percentage is " + percentage);
        System.out.println("the caluculation is completed");
    }

    void show() {
        System.out.println("----student details ----");
        System.out.println("name of student is " + studentname);
        System.out.println("roll number is " + rollno);
        System.out.println("s1 marks is " + s1);
        System.out.println("s2 marks is " + s2);
        System.out.println("s3 marks is " + s3);
        System.out.println("total marks is " + total);
        System.out.println("total percentage is " + percentage);
    }
}

public class StudentProgram {
    public static void main(String[] args) {
        Scanner sp = new Scanner(System.in);
    }
}

```

```

        Student st = new Student();
        int choice;

        do {
            System.out.println("
--- student utility program---");
            System.out.println("1.enter student details ");
            System.out.println("2.calc total and percentage");
            System.out.println("3.show result");
            System.out.println("4.exit");
            System.out.println("enter your choice");
            choice = sp.nextInt();

            switch(choice) {
                case 1:
                    st.studentdetails();
                    break;
                case 2:
                    st.calc();
                    break;
                case 3:
                    st.show();
                    break;
                case 4:
                    System.out.println("exited ");
                    break;
                default:
                    System.out.println("invalid choice");
            }
        } while (choice != 4);

        sp.close();
    }
}

```

## Sample Output

```
--- student utility program---
1.enter student details
2.calc total and percentage
3.show result
4.exit
enter your choice
1
enter the name of student
John Doe
enter th roll number
101
enter the firstsubject marks
85
enter the secondsubject marks
90
enter the thirdsubject marks
80

--- student utility program---
...
enter your choice
2
total is 255
percentage is 85.0
the caluculation is completed

--- student utility program---
...
enter your choice
3
----student details ----
name of student is John Doe
roll number is 101
s1 marks is 85
s2 marks is 90
s3 marks is 80
total marks is 255
total percentage is 85.0
```